Progress Report

Day 3 — Database Integration & Backend Infrastructure
Task Management MERN Stack Platform
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1. Day-3 Objective

The goal for Day 3 was to configure MongoDB Atlas and MySQL database connectivity, integrate both databases into the Express backend, and implement centralized database configuration architecture.

2. Tasks Completed

MongoDB Atlas Setup

- Created MongoDB Atlas project and cluster
- Configured database user and network access
- Generated Node.js connection string for backend integration
- Connected backend using Mongoose

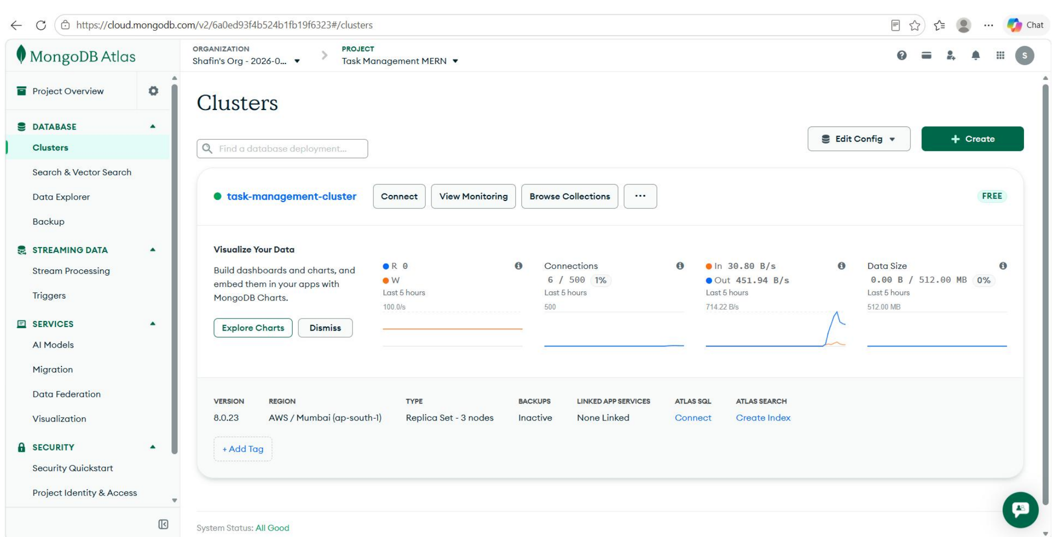

Figure 2.1 — MongoDB Atlas cluster configured successfully.

Local MySQL Setup

- Installed MySQL Community Server and MySQL Workbench
- Configured local MySQL server
- Created task_management_audit schema
- Verified local database connectivity

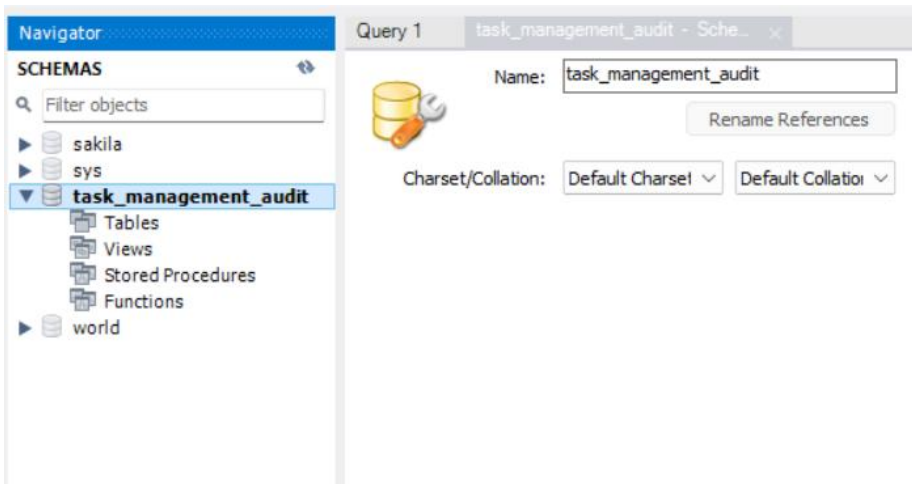

Figure 2.2 — Local MySQL schema created using MySQL Workbench.

Backend Database Configuration

- Created centralized database configuration modules:
  - config/mongo.js
  - config/mysql.js

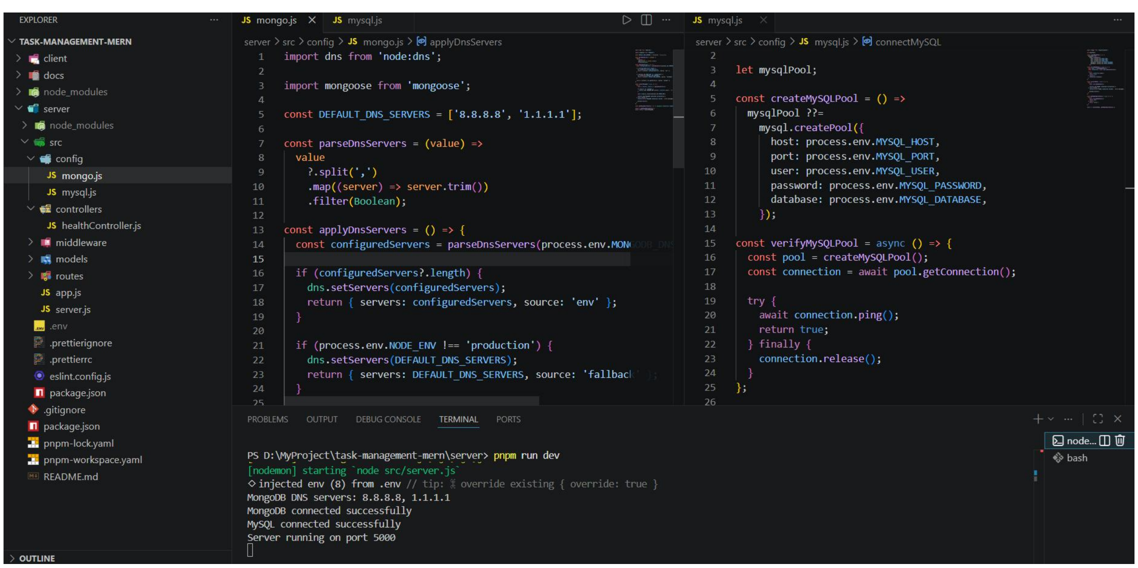

Figure 2.3 — Centralized database configuration modules inside backend architecture.

Backend Startup Flow

- Updated backend startup flow so database connections initialize before app.listen()
- Added startup logging for MongoDB and MySQL connections
- Configured backend to stop startup if database connection fails

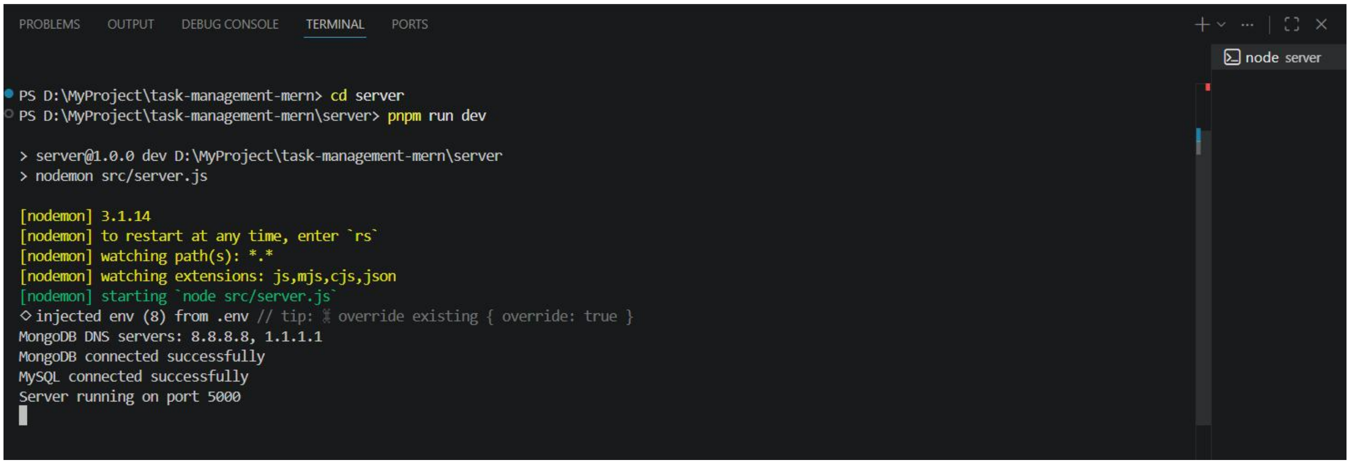

Figure 2.4 — Backend server started successfully after dual database connection.

Health Endpoint Integration

- Updated /api/health endpoint to return real database connection status
- Verified successful MongoDB and MySQL connection responses from browser

Current API response:

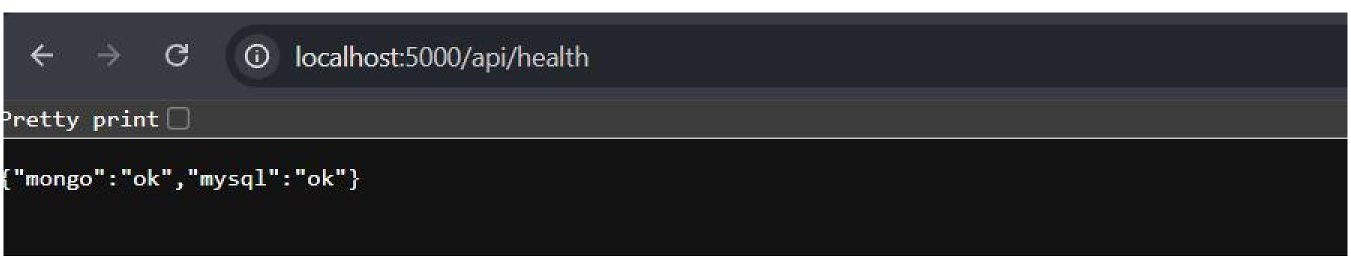

Figure 2.5 — Health endpoint returning real MongoDB and MySQL connection status.

Environment Configuration

- Configured backend .env structure for:

- MongoDB Atlas
- MySQL
- JWT secret
- backend port

- Separated sensitive credentials from source code

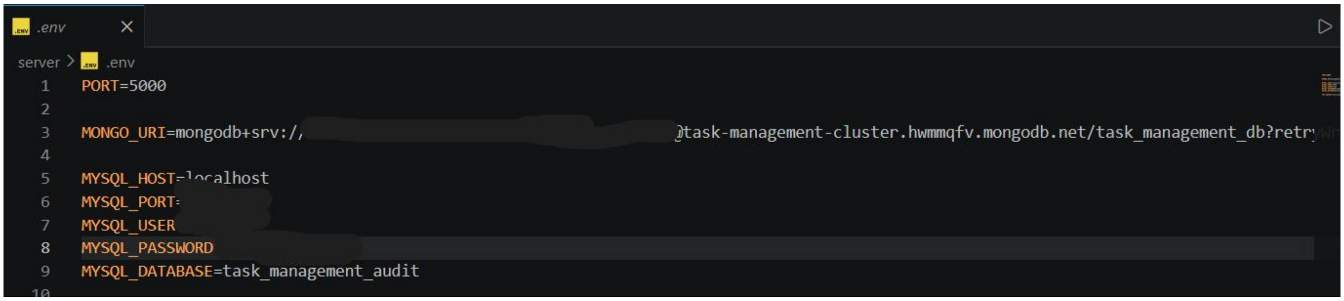

Figure 2.6 — Environment variable configuration for backend database integration.

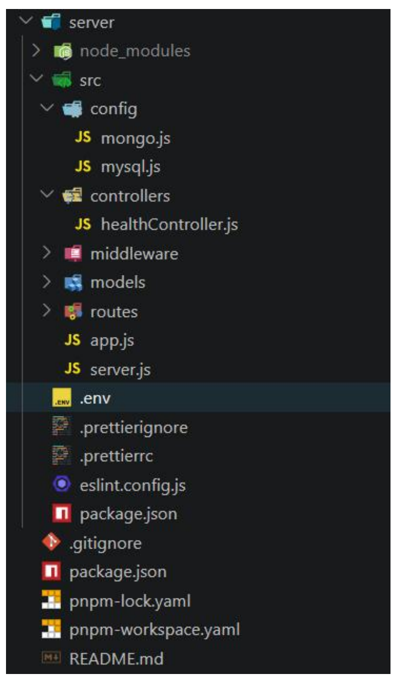

Figure 2.7 — Updated workspace structure after Day 3 backend infrastructure setup.

3. Challenges Faced

- MongoDB Atlas SRV DNS resolution initially failed on the local Windows environment and required DNS fallback handling during development

- MySQL connection pool initialization initially occurred before environment variables were loaded and required startup flow adjustment

- Database startup sequencing required restructuring so the server only starts after successful database connections

4. Next Steps (Day-4)

- Begin User schema planning using MongoDB
- Implement authentication structure using JWT
- Create authentication routes and controllers
- Prepare middleware for protected routes
- Begin Team and Task model planning

5. Conclusion

Day 3 focused on database integration and backend infrastructure setup. MongoDB Atlas and MySQL were successfully connected to the Express backend, centralized database configuration modules were implemented, and the API health endpoint now returns real database connection status.